

# Comparative Time Series Analysis and Forecasting of Mobile Network Traffic

This formative assignment serves to assess your programming and data analysis capabilities in time series analysis and forecasting. It is aligned with machine learning methods for sequential data, including classical statistical modeling and neural network-based approaches covered in the ML techniques I course.

**Note that the full data amounts to around 5 GBytes, hence it is suggested to start retrieving them well in advance in case of slow Internet connections.**

The assignment is about analyzing a dataset of mobile data traffic recorded in a real-world network that covers a large city. The dataset is organized in lines, each reporting traffic information for a specific combination of location and time, in the following format.

- Square id: the identifier of a given geographical area within the target city;
- Time Interval: a 10-minute time interval;
- Internet traffic activity: a measure of the mobile network traffic activity in “Square id” during “Time Interval”.

The dataset reports traffic information for 10,000 areas (organized in a 100×100 regular grid) during a continued period of two months. Using this dataset, you are required to complete the three tasks described below.

## Task 1: Data Handling & Memory Management

In the first task, you are required to demonstrate the ability to efficiently handle and process large-scale data. The dataset used in this assignment is approximately 5GB in size, and managing it effectively within available computational resources is an essential part of the task. You are expected to explore techniques for reducing memory usage and improving computational efficiency. This may include optimizing data types, reducing precision where appropriate, and restructuring the dataset to minimize memory overhead. You must:

- I. Describe the strategy used to load and process the dataset, including any techniques applied to manage memory usage.
- II. Explain any data reduction or transformation methods used and justify why these choices were appropriate.
- III. The chosen approach(es) must be justified, and its impact on memory usage should be clearly demonstrated. You must report memory usage before and after applying optimization techniques.
- IV. Provide a brief discussion of challenges encountered when working with the dataset and how they were addressed.
- V. Provide a brief description of the hardware and software setup used to process the data, including any limitations encountered and how they influenced the approach.

The ability to work with large datasets, make appropriate design decisions under resource constraints, and implement efficient data handling strategies is a key component of this assignment.

## Task 2: Exploratory Data Analysis/Data Characterization

In the second task, you have to perform a basic characterization of the data. Specifically, the following results should be produced as a minimum.

- I. A plot of the probability density function of the traffic in the target city, computed over 10,000 samples that each represent the total two-month traffic in one geographical area, and with an accompanying text

where you comment on the result (e.g., discussing the homogeneity or diversity of the distribution, and providing possible explanations for the observed behaviors)

- II. A figure of the time series of network traffic during the first two weeks at three areas, namely (i) the area with the highest total traffic during the two-month period, (ii) the area with Square id 4159, and (iii) the area with Square id 4556, with an accompanying text where you comment on the result (e.g., discussing the similarities or differences observed in the temporal dynamics of each area, and speculating on their causes).
- III. An analysis of stationarity of the time series, including rolling statistics and the Augmented Dickey-Fuller (ADF) test, with interpretation of the results;
- IV. A decomposition of the time series into trend, seasonal, and residual components, with discussion of periodic patterns observed in the data;
- V. Autocorrelation (ACF) and partial autocorrelation (PACF) plots, together with a brief explanation of what they reveal about temporal dependencies;
- VI. At least one spatial analysis of the dataset (e.g., a heatmap of traffic intensity across the grid), with discussion of spatial patterns;
- VII. A discussion of anomalies, outliers, or unusual behaviors observed in the dataset, and possible explanations.

Note: Your discussions should connect the plots to possible real-world explanations.

You are free (and encouraged) to provide additional results that explore any other properties of interest of the data. Supplementary figures and comments that are insightful will be considered positively during grading.

### Task 3: Forecasting Models

In the third task, you are asked to design, implement, and compare multiple algorithms (exactly 3 algorithms) for one-step-ahead prediction of network traffic in a single geographical area. The objective of this task is to explore both classical statistical approaches (one statistical approach) and two neural network-based methods for modeling sequential data. Formally, let us denote as  $x_a(t)$  the traffic observed at area  $a$  during time interval  $t$ . At each time  $t$ , the algorithm receives as input a history  $x_t$ , i.e., a vector of traffic values in past time intervals, up to  $t$  included. The algorithm shall then produce as output an estimate  $\overline{x}_a(t + 1)$  of future traffic at  $t + 1$  in area  $a$ . You shall run the algorithm to forecast traffic in the three geographical areas identified at the second item of Task 2, during the week from December 16 to 22. The same set of models must be applied consistently across all three geographical areas. Each model must be trained and evaluated independently on each of the three geographical areas. The following results should be produced as a minimum:

- I. A self-contained description of all proposed models, where each model is clearly described in terms of its structure, input representation, preprocessing steps, and training procedure
- II. Plots reporting the superposed time series of (i) the original traffic and (ii) the predicted traffic in the week from December 16 to 22, for each model and for each of the three geographical areas (a total of 9 plots);
- III. Three tables reporting the performance of all selected models for each of the three geographical areas, where each table presents the Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Squared Error (RMSE) for all models
- IV. Exact statistics on the training and execution time of each model, with details on the process used to compute such statistics and on the hardware on which they were recorded; The reported training and execution times should be representative of the model and may be obtained from one selected area or as an average across the three areas, provided the method used is clearly stated.

- V. Explanation text where you provide personal considerations on the design, performance, and possible margins for improvement of all models.
- VI. You must clearly describe the input representation used for each model, including the sequence length, preprocessing steps, and normalization techniques.
- VII. You must provide a comparative analysis of all models, discussing differences in predictive performance, training time, and suitability for the dataset. The identification of the best-performing model must be supported by both quantitative results and qualitative reasoning based on the observed data characteristics from Task 2
- VIII. You must also include a failure analysis by identifying at least one time period where the models perform poorly and explaining the possible causes based on the data patterns.

Submissions that describe only a single model or do not provide a complete comparison across all implemented models will be considered incomplete. You can provide additional results that complement those above, only if they are strictly needed to highlight some important properties of the models.

### The Dataset

The reference dataset for the homework was released by Telecom Italia Mobile (TIM) for a data analysis challenge. Use the links provided in the references section to download the data. A preliminary description of the data is provided in a paper published in Scientific Data by Barlacchi et al. [1]. While the paper presents the many datasets that were made available for the challenge, only two are relevant to the homework:

- the Telecommunications activity dataset for the city of Milan (i.e., data citation 5 in the paper), which contains mobile network traffic information;
- The Grid dataset for the city of Milan (i.e., data citation 2 in the paper), which describes the tessellation of space into the areas over which such information is aggregated.
- As detailed in [1], the network traffic is measured in terms of number of call detail records (CDR) generated by mobile Internet sessions. This is only a proxy for the actual traffic volume in bytes, which provides however a decent estimate for the spatiotemporal dynamics of mobile data traffic.
- You can download both datasets from the Harvard Dataverse repository, at [2] and [3], respectively. In the case of the Telecommunications activity dataset, only the three fields Square id, Time Interval, and Internet traffic activity are relevant to the homework, and all other fields can be ignored.
- **Note:** the full data amount to around 5 GBytes, hence it is suggested to start retrieving them well in advance in case of slow Internet connections.
- **Note:** there is an error in [1] about the ordering of the fields: the country code is the third field, and all others are shifted accordingly.

You may employ any tool that they deem appropriate to solve the tasks outlined above. This includes any programming language or combination of programming languages, libraries, open source code, or solutions previously proposed in the literature. For Task 3, you have complete freedom in the choice of approach, as well as on the parametrization of the algorithms; specifically, any definition of history  $x_t$  is allowed, for example, in terms of the number of past traffic values it comprises, and of their associated time intervals and areas. Note, however, that the week to be predicted, from December 16 to 22, shall be used for test only, and cannot be part of the data used for training or validation, when those apply.

### References

[1] Gianni Barlacchi, Marco De Nadai, Roberto Larcher, Antonio Casella, Cristiana Chitic, Giovanni Torrisi, Fabrizio Antonelli, Alessandro Vespignani, Alex Pentland, and Bruno Lepri, “A multi-source dataset of urban

life in the city of Milan and the Province of Trentino.” Sci Data 2, 150055 (2015).  
<https://doi.org/10.1038/sdata.2015.55>

[2] <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/EGZHFV>

[3] <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/QJWLFU>

## **Deliverable**

You are required to submit the following documents by the agreed deadline:

1. a Portable Document Format (PDF) document containing all text and figures detailed in Section I, which should be as concise as possible, and clearly cite all sources (libraries, open source code, web sources, scientific methods, research papers) used in the project;
2. a single and well-organized archive or GitHub repository with (i) all source code, (ii) the minimum amount of input data needed to run the prediction algorithm, and (iii) precise instructions on how to run the prediction algorithm on Windows, Linux, or macOS;
3. a recorded video demonstration (5–8 minutes) showing the implementation, key results, and explanations of at least one design decision and one limitation or failure case. The video link must be included in the report. You must provide explanations that are specific to your implementation and results. Generic or non-specific descriptions will be penalized.

## **Academic Integrity**

- You may use external resources, including AI tools, as support. However, all submitted work must reflect your own understanding and you must clearly state where and how AI was used in any part of the Assignment.
- The submitted code, report, and video will be used to assess your level of understanding. In the video, you must provide explanations that are specific to your implementation and results. Generic or non-specific descriptions will be penalized.
- You are expected to be able to explain all parts of your submission, including code, results, and design decisions.
- A subset of students will be randomly selected and invited for a short oral viva (defense), during which they will be asked to explain and justify aspects of your implementation, modeling choices, and results. The purpose of this is to assess individual understanding of the submitted work.